

22 15 00 - General Service Compressed-Air Systems

Acceptable Piping Materials;

Laboratory & Instrument Air Applications: Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing pre-washed, degreased and capped at both ends by manufacture. Fittings pre-washed, degreased and wrapped by the manufacturer. All joints brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to be prevent oxidation.

Alternate material for Shop Air Applications Only: Schedule 40 galvanized steel pipe with treaded 150 pound galvanized malleable iron fittings.

Alternate material for Shop Air Applications Only: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder.

Building (house) compressed air shall not be shared with the building pneumatic compressed air system for HVAC controls. Building (house) compressed air and HVAC pneumatic control air shall be separate stand-alone systems complete with their own dedicated compressors.

For laboratory and instrument air, provide oil filter with automatic drain, and refrigerated air drier. For shop air applications, confirm requirements with users.

Valving: Compressed air systems shall be provided with manual isolation valves at the following locations:

- At each floor where branch piping connects to a riser.
- At the compressor.
- At branch connections to mains.
- Upstream of any equipment connected to compressed air.

Shop air systems typically require quick couple disconnect fittings for connection of flexible hoses. Confirm requirements with users.

Equipment:

Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

Requirements:

Drain Lines: Daylight to a conspicuous location. Plumb to sewer.

Hammer Arrestors: Install in a maintenance-accessible location for.